A Comparative Study of Moral Framing in Real and LLM-Generated Partisan Climate Headlines

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Abstract

This study compares the moral framing of real and synthetic partisan climate coverage. Real climate news titles shared by Democrat- and Republican-leaning US audiences on Earth Day 2016 and 2017 (Media Cloud; $N = 362$ and 113) are set against a size-matched corpus of titles written by Claude Sonnet 4.6. Two results follow. Real Democrat and Republican titles are morally close: none of the five foundations of the extended Moral Foundations Dictionary separates them (all $|d| < 0.5$, none surviving correction), and the sixth—liberty, measured with the LibertyMFD lexicon—does not separate them robustly either. What distinguishes the two sides is topical vocabulary, not moral loading. The synthetic corpus does not reproduce this picture. It lifts care, authority and sanctity well above their real levels (care $d = 0.68$, $p < .001$) and opens significant partisan gaps in fairness and loyalty that the real data does not contain. An LLM asked to write partisan climate titles therefore over-moralises and over-polarises them relative to the real thing, which has direct consequences for studies that treat generated text as a stand-in for real discourse.

1 Introduction

Climate change is one of the most polarised issues in US politics, and a growing literature uses text analysis to describe how partisan media frame it. Within this project, an emotion-based analysis with EmoAtlas [2] reports that language models overstate the partisan signal found in real coverage. Emotion is only one layer of framing, however. In Entman's terms [1], framing is about which concepts and values an issue is tied to, and the standard cognitive account of the value layer is Moral Foundations Theory (MFT). MFT distinguishes the individualising foundations (care, fairness) from the binding ones (loyalty, authority, sanctity), and holds that the political left draws more on the former and the right more on the latter [3]. A sixth foundation, liberty, was proposed later [4]. This left–right contrast is the assumption behind moral-reframing appeals in climate communication [5, 6], yet it is rarely tested on the partisan discourse itself.

Two primary questions are addressed. First, how morally polarised is real partisan climate discourse, and does the MFT contrast actually appear in it? Second, does a size-matched, LLM-generated corpus reproduce whatever moral structure the real data has? The second question speaks to the use of language models as "silicon samples" of human groups, whose usefulness depends on their algorithmic fidelity [10]. The present study differs from the emotion analysis in what it measures (moral foundations rather than affect), in method (between-group tests with effect sizes), and in taking the real corpus, not the model, as its primary object.

2 Data and Methods

2.1 Corpora

Real titles were collected through the Media Cloud API by querying climate keywords (climate change/crisis/action/emergency, global warming) on 22 April 2016 and 2017 across Media Cloud's partisan-audience quintile collections, where an outlet's quintile reflects the partisanship of the audience that shares it. Following the project protocol, quintiles Q1–Q2 are treated as the Democrat-leaning group and Q4–Q5 as the Republican-leaning group, and remove duplicate titles. This gives Democrat $N = 362$ (178 in 2016, 184 in 2017) and Republican $N = 113$ (53, 60).

2.2 Corpus generation

Synthetic titles are a size-matched corpus (363/113) written by Claude Sonnet 4.6 (Anthropic) from a single fixed prompt per party. The prompt gave the date, the required count, the topical keywords above, the partisan alignment (what Democrat- or Republican-leaning voters would have shared), and an instruction to invent titles rather than retrieve real ones. The full prompt and data are available in the project repository [11].

2.3 Measures

Titles were lowercased, POS-tagged and lemmatised (NLTK), keeping content words. Moral content is scored with the extended Moral Foundations Dictionary (eMFD) [7], which gives each word continuous probabilities of expressing the five foundations; a title's score is the mean over its matched words. Titles are short (mean 3.4 matched words for real, 4.6–5.1 for synthetic), hence, a single measure is not solely relied upon. The sixth foundation is added using the validated LibertyMFD lexicon [8], scoring each title as the mean liberty valence over its matched words, and all three released versions of that lexicon are reported as a robustness check. Separately, weighted log-odds with an informative Dirichlet prior [9] recover each group's characteristic vocabulary, and the Jaccard overlap of the top-25 partisan terms measures how far the real and synthetic frames coincide. Group contrasts use two-sided permutation tests (2×10^4 resamples) with Cohen's d, 95% bootstrap confidence intervals, and Bonferroni correction across the five eMFD foundations ($\alpha = .01$).

3 Results

3.1 Real partisan discourse is barely moralised at the foundation level

Table 1 describes the real corpus. No eMFD foundation survives Bonferroni correction. More than that, the effects are bounded small: every 95% bootstrap confidence interval falls within $|d| < 0.5$, the largest being fairness ($d = 0.23$, CI $[-0.45, -0.02]$). The MFT contrast does not appear. Democrats are not more individualising—if anything the Republican-leaning corpus is marginally more so ($p = .051$)—the binding foundations do not differ ($p = .28$), and the individualising/binding ratio is nearly identical across parties (0.70 vs 0.73).

Table 1: Mean eMFD foundation probability by corpus, with the individualising (care+fairness) and binding (loyalty+authority+sanctity) composites.

	Real (Media Cloud)		**Synthetic (LLM)**	
Foundation	Dem	Rep	Dem	Rep
Care	.101	.107	.125	.123
Fairness	.088	.096	.091	.098
Loyalty	.092	.096	.090	.097
Authority	.091	.094	.102	.102
Sanctity	.088	.089	.093	.094
Individualising	.189	.203	.215	.222
Binding	.270	.279	.285	.293
I/B ratio	.700	.730	.755	.758

The sixth foundation behaves the same way: across the three LibertyMFD versions the real partisan difference ranges from $d = 0.10$ to $d = 0.34$ and reaches significance only in the smallest lexicon, hence it is not interpreted as a reliable effect. On every validated moral instrument applied, the two audiences look alike.

3.2 The partisan difference is topical, not moral

Where the foundations are quiet, vocabulary is not. The Democrat-audience frame is built around the treaty and its politics—*Paris, accord, policy, deal, Washington*—with some noise from the day's other news (*prince*, the musician who had just died). The Republican-audience frame turns on scepticism and opposition—*skeptic, wrong, liberal, protester*—with an economic thread (*cost*). This is a difference in what the two sides talk about, not in the moral register they use to talk about it, and it is exactly what the foundation scores miss.

Table 2: Fidelity diagnostics. Partisan gap p: is the Dem–Rep difference significant within each corpus? Inflation d: real-vs-synthetic effect size within party (negative = synthetic higher).

	partisan gap p		**inflation d**	
Foundation	real	synth	Dem	Rep
Care	.18	.69	−0.68	−0.45
Fairness	.03	**.002**	−0.10	−0.08
Loyalty	.24	**.002**	+0.09	−0.06
Authority	.32	.92	−0.40	−0.26
Sanctity	.68	.82	−0.26	−0.20

3.3 The LLM over-moralises and over-polarises

The synthetic corpus departs from the real one in two consistent ways (Table 2). First, it raises the overall moral load: care, authority and sanctity are all higher in synthetic than in real titles, for both parties (care $d = 0.68$ Dem / 0.45 Rep; authority 0.40/0.26; sanctity 0.26/0.20; all $p < .001$ except Republican authority, $p = .056$). Second, it sharpens the partisan contrast: the weak real fairness gap and the non-significant real loyalty gap both become significant in the synthetic data ($p = .002$), so the model produces a moral divide that the real corpus does not support. The two frames also share little vocabulary—the top-25 partisan terms overlap at Jaccard .09 (Democrat) and .02 (Republican). The model captures the topic of partisan climate coverage but neither its muted moral profile nor its real wording.

4 Discussion

The moral distance between Democrat- and Republican-audience climate titles is small. Neither the five eMFD foundations nor the liberty foundation reliably tells the two groups apart, so the premise behind moral-reframing communication—that the sides speak in different foundational registers [5, 6]—is not visible in the headlines they circulate. The partisan signal that does exist is topical: a treaty-and-policy frame on one side, a scepticism-and-cost frame on the other.

The synthetic corpus tells a different and, for present purposes, a more informative story. Benchmarked against real data, the model is reliably more moralised and more polarised than the discourse it imitates. It inflates three of the five foundations and manufactures partisan gaps in the other two. A plausible reading is that a model asked to write "what a partisan would share" produces the prototype of each side rather than its ordinary output, and prototypes are cleaner and more moralised than everyday headlines. Either way, the practical lesson is concrete: using generated text as a proxy for real partisan discourse [10] will overstate both how moral and how divided that discourse is, and synthetic corpora should be checked against real data before they are trusted.

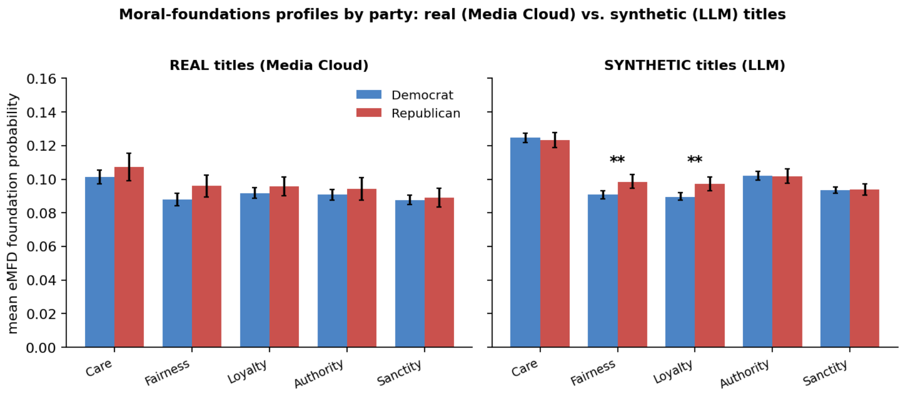

Figure 1: Mean eMFD foundation probabilities by party, with bootstrap 95% confidence intervals. In the real corpus (left) the Democrat and Republican profiles almost coincide; in the synthetic corpus (right) the partisan gap widens (fairness and loyalty reach significance, "**") and care, authority and sanctity sit well above their real levels.

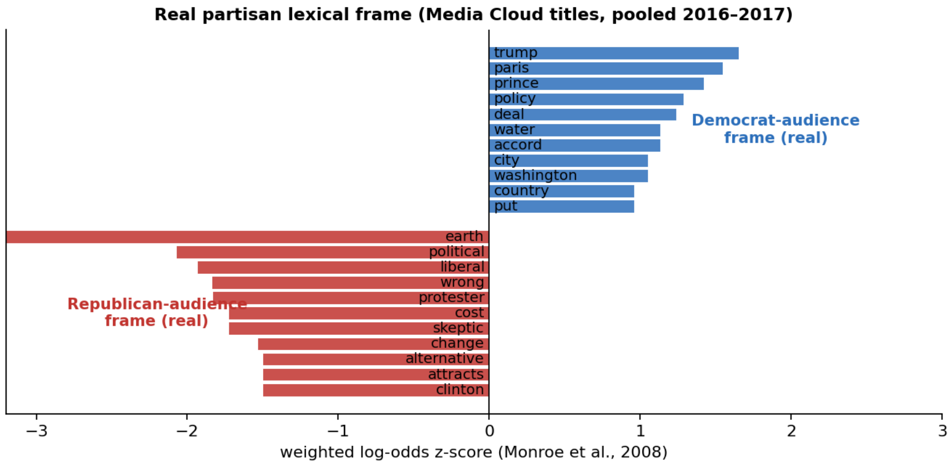

Figure 2: Real partisan vocabulary (weighted log-odds). This wording is almost disjoint from the synthetic corpus's (top-25 Jaccard $\leq .09$).

Limitations. The real corpus covers two single days and carries their topical noise (a celebrity death, the 2016 campaign), and the Republican sample is small ($N = 113$), so the null partisan result is reported as bounded, not proven; the inferential weight rests on the well-powered real-versus-synthetic comparisons. Titles are short, so eMFD leans on few words, and LibertyMFD's coverage is lower still (2–6 words per title), which is one reason its estimate is unstable across versions. The synthetic corpus is a single generation from one model and prompt, so the magnitudes reported here are specific to it.

5 Conclusion

This study demonstrates that real partisan climate discourse in media headlines is primarily distinguished by topical framing rather than fundamental moral divergence. Contrary to the assumptions of moral-reframing theories, Democrat- and Republican-leaning audiences circulate coverage with remarkably similar moral profiles. However, synthetic corpora generated by Large Language Models fail to capture this empirical reality, instead producing over-moralised and artificially polarised text. These findings issue a clear methodological warning for computational social science: while LLMs are powerful tools for text generation, treating them as unverified proxies for human groups risks amplifying polarisation effects that do not exist in the field. Future work should continue to establish rigorous empirical benchmarks before relying on synthetic data for political and social analysis.

References

[1] Entman, R.M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4), 51-58.

[2] Semeraro, A., Vilella, S., Improta, R., De Duro, E. S., Mohammad, S. M., Ruffo, G., & Stella, M. (2025). EmoAtlas: An emotional network analyzer of texts. Behavior Research Methods, 57, 77.

[3] Graham, J., Haidt, J., & Nosek, B.A. (2009). Liberals and conservatives rely on different sets of moral foundations. *Journal of Personality and Social Psychology*, 96(5), 1029-1046.

[4] Iyer, R., Koleva, S., Graham, J., Ditto, P., & Haidt, J. (2012). Understanding libertarian morality. *PLoS ONE*, 7(8), e42366.

[5] Feinberg, M., & Willer, R. (2013). The moral roots of environmental attitudes. *Psychological Science*, 24(1), 56-62.

[6] Wolsko, C., Ariceaga, H., & Seiden, J. (2016). Red, white, and blue enough to be green: moral framing effects on climate attitudes. *Journal of Experimental Social Psychology*, 65, 7-19.

[7] Hopp, F.R., Fisher, J.T., Cornell, D., Huskey, R., & Weber, R. (2021). The extended Moral Foundations Dictionary (eMFD). *Behavior Research Methods*, 53(1), 232-246.

[8] Araque, O., Gatti, L., & Kalimeri, K. (2022). LibertyMFD: A lexicon to assess the moral foundation of liberty. In Proceedings of the 2022 ACM Conference on Information Technology for Social Good (GoodIT '22), 154–160. ACM.

[9] Monroe, B.L., Colaresi, M.P., & Quinn, K.M. (2008). Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, 16(4), 372-403.

[10] Argyle, L.P., Busby, E.C., Fulda, N., Gubler, J.R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3), 337-351.

[11] Abazaj, I. (2026). A Comparative Study of Moral Framing in Real and LLM-Generated Partisan Climate Headlines. GitHub Repository, `https://github.com/ilvaabazaj-cell/A-Comparative-Study-of-Moral-Framing-in-Real-and-LLM-Generated-Partisan-Climate-Headlines`.